

Final Project Report

Vanishing Clouds: Mapping the Hidden Decline of Colombia's Páramo Ecosystems **Juan Sebastian Cortes**

For my final project, I created an interactive web map titled “Vanishing Clouds: Colombia’s Hidden Páramo Crisis.” The site is organized into six main sections: Páramos, Build a Páramo, Species, Threats, Urgency, and About, and it includes map layers such as páramo boundaries, species occurrences, elevation zones, agriculture pressure, fire alerts, and an urgency index. My main intention was to highlight how special the páramo is as an ecosystem in Colombia. We often talk about endemic species, but what fascinated me in this project is the idea that Colombia contains a remarkably concentrated and distinctive high-mountain ecosystem that feels almost “endemic” at the ecosystem scale. I wanted viewers to understand that the páramo is not just a remote landscape, but a rare ecological system tied to water regulation, biodiversity, and national environmental identity. My audience includes the general public, students, and anyone interested in conservation, but I also wanted the map to be meaningful for researchers and policymakers.

One of the biggest challenges in this project was not only collecting and combining different datasets but also adapting them for the web. I worked with multiple data sources and formats, including species records, páramo boundaries, elevation and climate layers, land-cover change, and fire data. These datasets had different spatial resolutions, symbology needs, and export limitations. A major technical challenge came from the fact that I was building the project for online viewing rather than only for desktop analysis. Many of the maps and layers that worked well on my laptop in ArcGIS Pro had to be adjusted to fit the formats supported by web layers. Raster layers often needed to be clipped, resampled, or simplified. Transparencies had to be adjusted, and in some cases, information had to be converted from raster to polygon so it could be published in a web-friendly vector format. This forced me to think not only as an analyst, but also as a translator between desktop GIS and web cartography.

The visualization decisions in the website were designed to guide the viewer through the páramo as both a place and a process. In the Páramos section, I focused on the overall geography of the ecosystem by showing the distribution of the páramo complexes. In Build a Páramo, I used environmental layers such as elevation and related conditions to explain the physical ingredients that make this ecosystem possible. In Species, I emphasized biodiversity and species occurrence patterns to show that these landscapes are biologically rich, not empty uplands. In Threats, I introduced agriculture pressure, fire alerts, and other human impacts to reveal how these ecosystems are being transformed. In Urgency, I plan to summarize those pressures into an index so viewers can quickly identify areas where conservation concern is greatest. Finally, the About section helps frame the broader purpose of the project. I think these decisions worked well because they create a clear narrative sequence: first understand what the páramo is, then how it forms, what lives there, what threatens it, and why conservation action matters. The site currently

presents 37 páramo complexes, 201 species records, 40 fire alerts, and 45 urgency zones, which helps communicate the scope of the project in a compact way.

Based on this visualization, I hope viewers come away with a stronger appreciation for how extraordinary Colombia's páramos are. I want them to understand that these ecosystems are not marginal spaces, but critical landscapes that support biodiversity, store and regulate water, and face growing pressure from land-use change and fire. More importantly, I hope the project changes the way people think about conservation by showing that it is not only individual endemic species that deserve attention, but also whole ecosystems whose uniqueness and fragility make them irreplaceable. By combining environmental conditions, biodiversity, and human pressure in one interactive map, the project encourages viewers to see the páramo as both ecologically exceptional and urgently in need of protection.

Data sources: The project integrates data from GBIF, INaturalist, NASA FIRMS, and MapBiomas, as listed on the website, along with supporting environmental and cartographic layers prepared in ArcGIS Pro for web display.

URL: <https://cortesju.github.io/vanishing-clouds/>